# ROBUST SIGNAL PROCESSING FOR WIRELESS SYSTEMS

**Çağrı Şükrü ÖZDOĞAN**

**İzmir Institute of Technology**
**October 2026**

# ROBUST SIGNAL PROCESSING FOR WIRELESS SYSTEMS

**A Thesis Submitted to**
**the Graduate School of Engineering and Sciences of**
**İzmir Institute of Technology**
**in Partial Fulfillment of the Requirements for the Degree of**

**DOCTOR OF PHILOSOPHY**

**in Electrical and Electronics Engineering**

**by**
**Çağrı Şükrü ÖZDOĞAN**

**October 2026**
**İZMİR**

We approve the thesis of Çağrı Şükrü ÖZDOĞAN

**Examining Committee Members:**

_____

**Prof. Dr. Committee MEMBER 1**
Electrical Engineering, Example University

_____

**Prof. Dr. Committee MEMBER 2**
Electrical Engineering, Example University

_____

**Prof. Dr. Committee MEMBER 3**
Electrical Engineering, Example University

_____

**Prof. Dr. Committee MEMBER 4**
Electrical Engineering, Example University

_____

**Prof. Dr. Committee MEMBER 5**
Electrical Engineering, Example University

10 October 2026

_____

**Prof. Dr. İlknur ÇİÇEK**
Supervisor
Electrical and Electronics Engineering,
IZTECH

_____

**Assoc. Prof. Gökhan IŞIK**
Co-Supervisor
Electrical and Electronics Engineering,
IZTECH

_____

**Prof. Dr. Ayşe YILMAZ**
Head of the Department
Electrical and Electronics Engineering

_____

**Prof. Dr. Deniz KAYA**
Dean of the Graduate School of
Engineering and Sciences

# PREFACE

This thesis combines three related articles.

# ACKNOWLEDGMENTS

I thank my supervisor and colleagues for their support.

# ABSTRACT

ROBUST SIGNAL PROCESSING FOR WIRELESS SYSTEMS

This verification specimen studies reproducible signal processing in wireless systems. The model compares measured observations with theoretical predictions and discusses limitations. The proposed method improves robustness while preserving computational efficiency. These sentences are deliberately long enough to test line wrapping, justification and paragraph indentation across the editor, exported PDF and Microsoft Word.

# ÖZET

## KABLOSUZ SİSTEMLER İÇİN DAYANIKLI SİNYAL İŞLEME

Bu doğrulama örneğinde kablosuz sistemlerde sinyal işleme incelenmektedir. Ölçüm sonuçları kuramsal tahminlerle karşılaştırılmış; yöntem, bulgular ve sınırlılıklar açıklanmıştır. Türkçe karakterler: ÇĞİÖŞÜ çğıöşü.

# TABLE OF CONTENTS

PREFACE..... iii
ACKNOWLEDGMENTS..... iv
ABSTRACT..... v
ÖZET..... vi
SYMBOLS AND ABBREVIATIONS..... viii
CHAPTER 1 ..... 1
INTRODUCTION
CHAPTER 2..... 2
ARTICLE 1
CHAPTER 3..... 3
ARTICLE 2
CHAPTER 4..... 4
ARTICLE 3
CHAPTER 5 ..... 5
CONCLUSIONS AND RECOMMENDATIONS
REFERENCES..... 6
APPENDIX A..... 7

# SYMBOLS AND ABBREVIATIONS

SNR: Signal-to-noise ratio

# CHAPTER 1

# INTRODUCTION

This verification specimen studies reproducible signal processing in wireless systems. The model compares measured observations with theoretical predictions and discusses limitations. The proposed method improves robustness while preserving computational efficiency. These sentences are deliberately long enough to test line wrapping, justification and paragraph indentation across the editor, exported PDF and Microsoft Word.

# CHAPTER 2

# ARTICLE 1

Publication citation: Example article 1, journal, volume, pages, DOI (test data).

This verification specimen studies reproducible signal processing in wireless systems. The model compares measured observations with theoretical predictions and discusses limitations. The proposed method improves robustness while preserving computational efficiency. These sentences are deliberately long enough to test line wrapping, justification and paragraph indentation across the editor, exported PDF and Microsoft Word.

# CHAPTER 3

# ARTICLE 2

Publication citation: Example article 2, journal, volume, pages, DOI (test data).

This verification specimen studies reproducible signal processing in wireless systems. The model compares measured observations with theoretical predictions and discusses limitations. The proposed method improves robustness while preserving computational efficiency. These sentences are deliberately long enough to test line wrapping, justification and paragraph indentation across the editor, exported PDF and Microsoft Word.

# CHAPTER 4

# ARTICLE 3

Publication citation: Example article 3, journal, volume, pages, DOI (test data).

This verification specimen studies reproducible signal processing in wireless systems. The model compares measured observations with theoretical predictions and discusses limitations. The proposed method improves robustness while preserving computational efficiency. These sentences are deliberately long enough to test line wrapping, justification and paragraph indentation across the editor, exported PDF and Microsoft Word.

# CHAPTER 5

# CONCLUSIONS AND RECOMMENDATIONS

This verification specimen studies reproducible signal processing in wireless systems. The model compares measured observations with theoretical predictions and discusses limitations. The proposed method improves robustness while preserving computational efficiency. These sentences are deliberately long enough to test line wrapping, justification and paragraph indentation across the editor, exported PDF and Microsoft Word.

# REFERENCES

Smith, A. Signal Processing Methods; Example Press: İzmir, 2026.

# APPENDIX A

Supplementary measurement protocol.

# VITA

Çağrı Şükrü ÖZDOĞAN studied electrical engineering and researches signal processing.